

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO4: Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 10

Total Marks:100

Annexure A

Scenario-Based Assessment

Code of Conduct:

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VIVETHA B

RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

NOTE:

For every question, answer the following:

- Identify the numerical and binary features used for KNN classification.
- Calculate the **Euclidean distance** between the new sample and all training samples
- Calculate the **Manhattan distance** between the new sample and all training samples.
- Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.
- Calculate the **Hamming distance** between the new sample and all training samples using binary features.
- Identify the **3 nearest neighbours** separately for each distance measure.
- Predict the class label using majority voting for each distance measure.
- Compare the results and explain which distance measure is most suitable for the given scenario.

Total = 100 Marks

Question 1: Student Pass or Fail Prediction

A college wants to predict whether a student will Pass or Fail based on academic and activity-related details.

Student	Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done	Result
S1	2	60	0	0	0	Fail
S2	3	65	1	0	0	Fail
S3	6	80	1	1	1	Pass
S4	7	85	1	1	1	Pass
S5	4	70	1	0	1	Fail
S6	8	90	1	1	1	Pass

A new student has the following details:

Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done
5	75	1	1	0

Using KNN with $K = 3$, classify the new student as Pass or Fail using Euclidean, Manhattan, Minkowski, and Hamming distance.

ANSWER

- Identify the numerical and binary features used for KNN classification.

In this KNN classification problem, the input features are divided into numerical features and binary features based on the type of values they contain.

Numerical Features

Numerical features are features whose values represent measurable quantities.

The dataset contains:

Numerical Feature	Description	Example Values
Study Hours	Number of hours the student studies	2, 3, 4, 6, 7, 8
Attendance %	Percentage of attendance	60, 65, 70, 80, 85, 90

Therefore:

Numerical Features = Study Hours, Attendance %

For the new student:

- Study Hours = 5
- Attendance = 75%

2. Binary Features

Binary features have only two possible values, generally 0 and 1.

The dataset contains:

Binary Feature	Meaning of 0	Meaning of 1
Assignment Submitted	Not submitted	Submitted
Lab Participation	Did not participate	Participated
Revision Done	No revision	Revision done

Therefore:

Binary Features = Assignment Submitted, Lab Participation, Revision Done

For the new student:

- Assignment Submitted = 1
- Lab Participation = 1
- Revision Done = 0

Target/Class Label

The Result is not an input feature. It is the target/class label that KNN needs to predict.

Possible classes are:

Pass, Fail

Complete Feature Representation

Thus, each student is represented using 5 input features:

$X = (\text{Study Hours}, \text{Attendance}\%, \text{Assignment}, \text{Lab}, \text{Revision})$

The new student's feature vector is:

$X = (5, 75, 1, 1, 0)$

Final Answer

Feature	Type	New Student
Study Hours	Numerical	5
Attendance %	Numerical	75
Assignment Submitted	Binary	1
Lab Participation	Binary	1

Revision Done	Binary	0
Result	Target/Class	To be predicted

Therefore:

Numerical Features: Study Hours, Attendance % Binary Features: Assignment Submitted, Lab Participation, Revision Done Target Variable: Result (Pass/Fail)

b) Calculate the **Euclidean distance** between the new sample and all training samples

The **Euclidean distance** measures the straight-line distance between the new student and each training student.

Given New Student

$X=(5,75,1,1,0)$

where:

- Study Hours = 5
- Attendance = 75
- Assignment = 1
- Lab = 1
- Revision = 0

(1)

col-ATT

(b)

Euclidean Distance Formula

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$$d = (\text{Studyhours}_1 - \text{Studyhours}_2)^2 + (\text{Attendance}_1 - \text{Attendance}_2)^2 + (\text{Lab}_1 - \text{Lab}_2)^2 + (\text{Revision}_1 - \text{Revision}_2)^2$$

1. Distance b/w New student and S1

$$S1 = (2, 60, 0, 0, 0)$$

$$x = (5, 75, 1, 1, 0)$$

$$d(x, S1) = \sqrt{(5-2)^2 + (75-60)^2 + (1-0)^2 + (1-0)^2 + (0-0)^2}$$

$$= \sqrt{3^2 + 15^2 + 1^2 + 1^2 + 0^2}$$

$$= \sqrt{9 + 225 + 1 + 1}$$

$$= \sqrt{236}$$

$$d(x, S1) = 15.362$$

2. Distance b/w New student and S2

$$S2 = (3, 65, 1, 0, 0)$$

$$d(x, S2) = \sqrt{(5-3)^2 + (75-65)^2 + (1-1)^2 + (1-0)^2 + (0-0)^2}$$

$$= \sqrt{2^2 + 10^2 + 0^2 + 1^2 + 0^2}$$

$$= \sqrt{4 + 100 + 0 + 1 + 0}$$

$$= \sqrt{105}$$

$$d(x, S2) = 10.247$$

3. Distance b/w New student and S3

$$S_3 = (6, 80, 1, 1, 1)$$

$$\begin{aligned} d(x, S_3) &= \sqrt{(5-6)^2 + (75-80)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2} \\ &= \sqrt{(-1)^2 + (-5)^2 + 0^2 + 0^2 + (-1)^2} \\ &= \sqrt{1 + 25 + 0 + 0 + 1} \\ &= \sqrt{27} \end{aligned}$$

$$d(x, S_3) = 5.196$$

4. Distance b/w New student and S4

$$S_4 = (7, 85, 1, 1, 1)$$

$$\begin{aligned} d(x, S_4) &= \sqrt{(5-7)^2 + (75-85)^2 + (-1-1)^2 + (1-1)^2 + (0-1)^2} \\ &= \sqrt{(-2)^2 + (-10)^2 + 0^2 + 0^2 + (-1)^2} \\ &= \sqrt{4 + 100 + 0 + 0 + 1} \\ &= \sqrt{105} \end{aligned}$$

$$d(x, S_4) = 10.247$$

5. Distance b/w New student and S5

$$S_5 = (4, 70, 1, 0, 1)$$

$$\begin{aligned} d(x, S_5) &= \sqrt{(5-4)^2 + (75-70)^2 + (1-1)^2 + (1-0)^2 + (0-1)^2} \\ &= \sqrt{1^2 + 5^2 + 0^2 + 1^2 + (-1)^2} \\ &= \sqrt{1 + 25 + 0 + 1 + 1} \\ &= \sqrt{28} \end{aligned}$$

$$d(x, S_5) = 5.292$$

6. Distance b/w new student and S6

$$S6 = (8, 90, 1, 1, 1)$$

$$d(x, S6) = \sqrt{(5-8)^2 + (75-90)^2 + (-1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= \sqrt{(-3)^2 + (-15)^2 + 0^2 + 0^2 + (-1)^2}$$

$$= \sqrt{9 + 225 + 0 + 0 + 1}$$

$$= \sqrt{235}$$

$$d(x, S6) = 15.330$$

Distance Ranking

From smallest to largest:

1. **S3 = 5.196 → Pass**
2. **S5 = 5.292 → Fail**
3. **S2 = 10.247 → Fail**
4. **S4 = 10.247 → Pass**
5. **S6 = 15.330 → Pass**
6. **S1 = 15.362 → Fail**

Since **K = 3**, the nearest neighbours are:

S3, S5, S2

There is a tie between **S2 and S4** for the third-nearest position, since both have distance **10.247**. Using S2 as the tie-break:

- S3 → Pass
- S5 → Fail
- S2 → Fail

Therefore, the Euclidean KNN prediction is:

FAIL

c) Calculate the **Manhattan distance** between the new sample and all training samples.

New Student

The new student's feature vector is:

$$X=(5,75,1,1,0)$$

where:

- Study Hours = 5
- Attendance = 75%
- Assignment Submitted = 1
- Lab Participation = 1
- Revision Done = 0

(c) Manhattan Distance formula

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

1. Distance b/w New student and S_1

$$S_1 = (2, 60, 0, 0, 0)$$

$$\begin{aligned} d(x, S_1) &= |5-2| + |75-60| + |1-0| + |1-0| + |0-0| \\ &= 3 + 15 + 1 + 1 \end{aligned}$$

$$d(x, S_1) = 20$$

2. Distance b/w New student and S_2

$$S_2 = (3, 65, 1, 0, 0)$$

$$\begin{aligned} d(x, S_2) &= |5-3| + |75-65| + |1-1| + |1-0| \\ &= 2 + 10 + 0 \end{aligned}$$

$$d(x, S_2) = 13$$

3) Distance b/w New student and S3

$$S_3 = (6, 20, 1, 1, 1)$$

$$d(x, S_3) = |5-6| + |75-20| + |1-1| + |1-1| + |0-1|$$
$$= 1+5+1$$

$$d(x, S_3) = 7$$

4) Distance b/w New student and S4

$$S_4 = (7, 25, 1, 1, 1)$$

$$d(x, S_4) = |5-7| + |75-25| + |1-1| + |1-1| + |0-1|$$
$$= 2+10+1$$

$$d(x, S_4) = 13$$

5) Distance b/w New student and S5

$$S_5 = (4, 20, 1, 0, 1)$$

$$d(x, S_5) = |5-4| + |75-20| + |1-1| + |1-0| + |0-1|$$
$$= 1+5+1+1$$

$$d(x, S_5) = 8$$

6) Distance b/w New student and S6

$$S_6 = (8, 90, 1, 1, 1)$$

$$d(x, S_6) = |5-8| + |75-90| + |1-1| + |1-1| + |0-1|$$
$$= 3+15+1$$

$$d(x, S_6) = 19$$

Distance Ranking

Arrange the distances from smallest to largest:

1. **S3 = 7 → Pass**
2. **S5 = 8 → Fail**
3. **S2 = 13 → Fail**
4. **S4 = 13 → Pass**
5. **S6 = 19 → Pass**
6. **S1 = 20 → Fail**

Since:

$K=3$

the three nearest neighbours are:

S3, S5, S2

There is a **tie between S2 and S4** for the third-nearest position because both have Manhattan distance 13. Using S2 as the tie-breaking choice:

- S3 → Pass
- S5 → Fail
- S2 → Fail

Majority Voting

Pass=1 Fail=2

Therefore, using Manhattan distance:

Predicted Class = FAIL

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) Minkowski Distance with $p=3$

formula

$$d(x, y) = \left(\sum |x_i - y_i|^3 \right)^{1/3}$$

$$x = (5, 25, 1, 1, 0)$$

$$s_1: (2, 60, 0, 0, 0)$$

$$[(3)^3 + (15)^3 + (1)^3 + (1)^3 + (0)^3]^{1/3}$$

$$= 15.038$$

$$s_2: (3, 65, 1, 0, 0)$$

$$[(2)^3 + (10)^3 + (0)^3 + (1)^3 + (0)^3]^{1/3}$$

$$= 10.030$$

$$s_3: (6, 20, 1, 1, 1)$$

$$[(1)^3 + (5)^3 + (0)^3 + (0)^3 + (1)^3]^{1/3}$$

$$= 5.027$$

$$s_4: (7, 85, 1, 1, 1)$$

$$[(2)^3 + (10)^3 + (0)^3 + (0)^3 + (1)^3]^{1/3}$$

$$= 10.030$$

$$s_5: (4, 70, 1, 0, 1)$$

$$[(1)^2 + (5)^3 + (0)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 5.040$$

$$s_6: (8, 90, 1, 1, 1)$$

$$[(3)^3 + (15)^3 + (0)^3 + (0)^3 + (1)^3]^{1/3}$$

$$= 15.038$$

Ranking from nearest to farthest

1. **S3 = 5.027 → Pass**
2. **S5 = 5.040 → Fail**
3. **S2 = 10.030 → Fail**
4. **S4 = 10.030 → Pass**
5. **S6 = 15.037 → Pass**
6. **S1 = 15.038 → Fail**

Since $K=3$, the nearest 3 are:

S3, S5, S2

Majority Voting

Neighbour	Class
-----------	-------

S3	Pass
----	------

S5	Fail
----	------

S2	Fail
----	------

Pass=1,Fail=2

Therefore:

Minkowski Distance Prediction = FAIL

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, we use **only the binary features**:

1. Assignment Submitted
2. Lab Participation
3. Revision Done

(a) Hamming Distance using Binary Features

$$X = (1, 1, 0)$$

Formula

$d_H(x, y)$ = No. of positions where the binary values are different

$$S_1: (0, 0, 0)$$

$$\begin{aligned} d_H &= |1-0| + |1-0| + |0-0| \\ &= 1 + 1 + 0 \end{aligned}$$

$$d_H(S_1) = 2$$

$$S_2: (1, 0, 0)$$

$$\begin{aligned} d_H &= |1-1| + |1-0| + |0-0| \\ &= 0 + 1 + 0 \end{aligned}$$

$$d_H(S_2) = 1$$

$$S_3: (1, 1, 1)$$

$$\begin{aligned} d_H &= |1-1| + |1-1| + |0-1| \\ &= 0 + 0 + 1 \end{aligned}$$

$$d_H(S_3) = 1$$

$$S_4: (1, 1, 1)$$

$$\begin{aligned} d_H &= |1-1| + |1-1| + |0-1| \\ &= 0 + 0 + 1 \end{aligned}$$

$$d_H(S_4) = 1$$

$$S_5: (1, 0, 1)$$

$$= 0 + 1 + 1$$

$$d_H(S_5) = 2$$

$$S_6: (1, 1, 1)$$

$$= 0 + 0 + 1$$

$$d_H(S_6) = 1$$

f) Identify the **3 nearest neighbours** separately for each distance measure

3 Nearest Neighbours

The minimum distance is:

1

Four students have distance 1:

S2, S3, S4, S6

There is therefore a **tie** for the 3 nearest neighbours.

Using the dataset order, select:

S2, S3, S4

Their classes are:

- S2 → Fail
- S3 → Pass
- S4 → Pass

Majority Voting

Pass=2 Fail=1

Therefore:

Hamming Prediction = PASS

g) Predict the class label using majority voting for each distance measure.

For **K = 3**, we select the 3 nearest neighbours for each distance measure and count their class labels.

1. Euclidean Distance

3 nearest neighbours:

- S3 → Pass
- S5 → Fail
- S2 → Fail

Pass=1, Fail=2 Prediction = FAIL

2. Manhattan Distance

3 nearest neighbours:

- S3 → Pass

- S5 → Fail
- S2 → Fail

Pass=1,Fail=2 Prediction = FAIL

3. Minkowski Distance (p=3)

3 nearest neighbours:

- S3 → Pass
- S5 → Fail
- S2 → Fail

Pass=1,Fail=2 Prediction = FAIL

4. Hamming Distance

Using the tie-breaking selection:

- S2 → Fail
- S3 → Pass
- S4 → Pass

Pass=2,Fail=1 Prediction = PASS

h) Compare the results and explain which distance measure is most suitable for the given scenario.

Explanation

- **Euclidean, Manhattan and Minkowski** all predict **Fail**.
- **Hamming** predicts **Pass**, because it considers only the binary features.
- For this problem, Hamming alone is **not the most suitable**, because it ignores important numerical features such as **Study Hours and Attendance %**.
- Euclidean, Manhattan and Minkowski use **all five features**, but Attendance % has a much larger scale and can dominate the distance.

Most Suitable Distance Measure

For this student dataset, **Euclidean distance with feature normalization/standardization** is a suitable choice because it considers both numerical and binary features.

Before applying KNN, numerical features should be normalized so that Attendance % does not dominate the calculation.

Most Suitable: Euclidean Distance after Feature Scaling

Python Code for the Graph

```
import matplotlib.pyplot as plt
```

```
distance_measures = [
```



```

"Euclidean",
"Manhattan",
"Minkowski (p=3)",
"Hamming"]
predicted_classes = ["Fail", "Fail", "Fail", "Pass"]
prediction_values = [0, 0, 0, 1]

plt.figure(figsize=(9, 5))

bars = plt.bar(distance_measures, prediction_values)

plt.xlabel("Distance Measure")

plt.ylabel("Predicted Class")

plt.title("KNN Prediction Comparison (K = 3)")

plt.yticks([0, 1], ["Fail", "Pass"])

plt.ylim(-0.2, 1.2)

for bar, label in zip(bars, predicted_classes):

    plt.text(

        bar.get_x() + bar.get_width() / 2,

        bar.get_height() + 0.03,

        label,

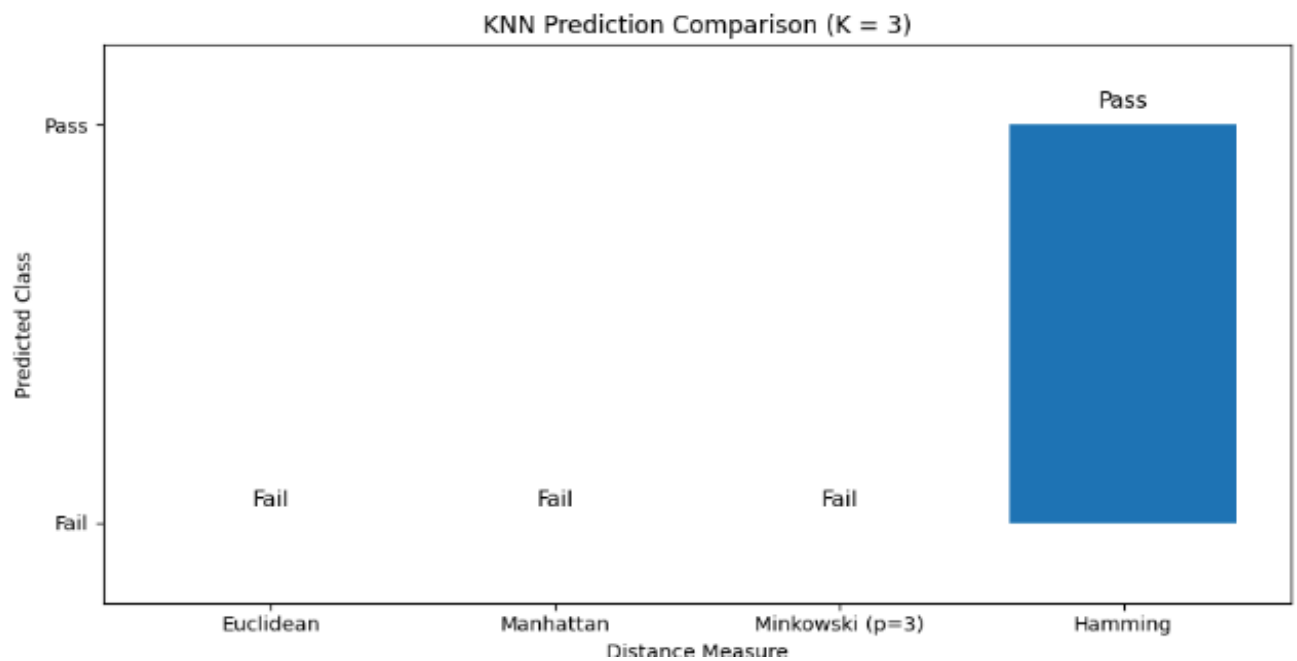
        ha="center",

        fontsize=11 )

plt.tight_layout()

plt.show()

```



Question 2: Online Customer Purchase Prediction

An e-commerce company wants to predict whether a customer will Buy or Not Buy a product.

Customer	Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer	Class
C1	10	3	0	0	0	Not Buy
C2	15	4	1	0	0	Not Buy
C3	25	7	1	1	1	Buy
C4	30	8	1	1	1	Buy
C5	18	5	0	1	0	Not Buy
C6	28	9	1	1	1	Buy

A new customer has:

Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer
22	6	1	1	0

Using KNN with K = 3, classify the new customer as Buy or Not Buy using all four distance measures.

a) Identify the numerical and binary features used for KNN classification.

In this KNN problem, the input features are divided into **numerical** and **binary** features.

1. Numerical Features

These features contain measurable numerical values:

- **Browsing Time**
- **Items Viewed**

Numerical Features = Browsing Time, Items Viewed

For the new customer:

Browsing Time=22 Items Viewed=6

2. Binary Features

These features have values **0 or 1**:

- **Coupon Used**
- **Added to Cart**
- **Returning Customer**

Binary Features = Coupon Used, Added to Cart, Returning Customer

For the new customer:

(1,1,0)

3. Target/Class Label

The **Class** column is the target variable to be predicted.

Possible classes:

Buy, Not Buy

b) Calculate the **Euclidean distance** between the new sample and all training samples

2) b) Euclidean Distance

$$x = (22, 6, 1, 1, 0)$$

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

1. Distance from c1

$$(10, 3, 0, 0, 0)$$

$$d(x, c_1) = \sqrt{(22-10)^2 + (6-3)^2 + (1-0)^2 + (1-0)^2 + (0-0)^2}$$

$$d(x, c_1) = 12.450$$

2. Distance from c2

$$(15, 4, 1, 0, 0)$$

$$d(x, c_2) = \sqrt{(22-15)^2 + (6-4)^2 + (1-1)^2 + (1-0)^2 + (0-0)^2}$$

$$= 7.348$$

3. Distance from c3

$$(25, 7, 1, 1, 1)$$

$$d(x, c_3) = \sqrt{(22-25)^2 + (6-7)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= 3.317$$

4. Distance from c4

$$(30, 8, 1, 1, 1)$$

$$d(x, c_4) = \sqrt{(22-30)^2 + (6-8)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= 8.307$$

5. Distance from c5

$$d(x, c_5) = \sqrt{(22-18)^2 + (6-5)^2 + (1-0)^2 + (1-1)^2 + (0-0)^2}$$

$$= 4.243$$

6. Distance from c6 (28, 9, 1, 1, 1)

$$d(x, c_6) = 6.782$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(c) Manhattan Distance

$$x = (22, 6, 1, 1, 0)$$

formula

$$d(x, y) = \sum |x_i - y_i|$$

calculation

(1) $|22 - 10| + |6 - 3| + |1 - 0| + |1 - 0| + |0 - 0| = 12 + 3 + 1 + 1 = 17$

(2) $|22 - 15| + |6 - 4| + |1 - 1| + |1 - 0| + |0 - 0| = 7 + 2 + 0 + 1 + 0 = 10$

(3) $|22 - 25| + |6 - 7| + |1 - 1| + |1 - 1| + |0 - 1| = 3 + 1 + 0 + 0 + 1 = 5$

(4) $|22 - 30| + |6 - 8| + |1 - 1| + |1 - 1| + |0 - 1| = 8 + 2 + 0 + 0 + 1 = 11$

(5) $|22 - 18| + |6 - 5| + |1 - 0| + |1 - 1| + |0 - 1| = 4 + 1 + 1 + 1 = 7$

(6) $|22 - 28| + |6 - 9| + |1 - 1| + |1 - 1| = 6 + 3 + 1 = 10$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) Minkowski Distance with $p = 3$

formula

$$d(x, y) = (\sum |x_i - y_i|^p)^{1/p}$$

$$x = (22, 6, 1, 1, 0)$$

(1) $[(12)^3 + (3)^3 + (1)^3 + (1)^3 + (0)^3]^{1/3} = \sqrt[3]{1757} = 12.066$

(2) $[(7)^3 + (2)^3 + (0)^3 + (1)^3 + (0)^3]^{1/3} = \sqrt[3]{352} = 7.061$

(3) $[(3)^3 + (1)^3 + (0)^3 + (1)^3]^{1/3} = \sqrt[3]{29} = 3.072$

(4) $[(8)^3 + (2)^3 + (0)^3 + (1)^3]^{1/3} = \sqrt[3]{521} = 8.044$

(5) $[(4)^3 + (1)^3 + (1)^3 + (0)^3 + (0)^3]^{1/3} = \sqrt[3]{66} = 4.041$

(6) $[(6)^3 + (3)^3 + (0)^3 + (0)^3 + (1)^3]^{1/3} = \sqrt[3]{244} = 6.246$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

Binary features:

- Coupon Used
- Added to Cart
- Returning Customer

New customer:

$X=(1,1,0)$

Formula

dH =Number of different binary positions

Customer	Binary Features	Hamming Distance	Class
C1	(0,0,0)	2	Not Buy
C2	(1,0,0)	1	Not Buy
C3	(1,1,1)	1	Buy
C4	(1,1,1)	1	Buy
C5	(0,1,0)	1	Not Buy
C6	(1,1,1)	1	Buy

There is a tie at distance 1. Using dataset order, select:

C2, C3, C4

f) Identify the **3 nearest neighbours** separately for each distance measure.

3 Nearest Neighbours

Distance	1st	2nd	3rd
Euclidean	C3 (3.317)	C5 (4.243)	C6 (6.782)
Manhattan	C3 (5)	C5 (6)	C2/C6 (10)
Minkowski $p=3$	C3 (3.072)	C5 (4.041)	C6 (6.246)
Hamming	C2 (1)	C3 (1)	C4 (1)

g) Predict the class label using majority voting for each distance measure.

Euclidean

C3 → Buy
C5 → Not Buy
C6 → Buy

Buy=2,NotBuy=1 Prediction = BUY

Manhattan

Using C2 as the tie-break:

C3 → Buy
C5 → Not Buy
C2 → Not Buy

Buy=1,NotBuy=2 Prediction = NOT BUY

Minkowski p=3

C3 → Buy
C5 → Not Buy
C6 → Buy

Buy=2,NotBuy=1 Prediction = BUY

Hamming

C2 → Not Buy
C3 → Buy
C4 → Buy

Buy=2,NotBuy=1 Prediction = BUY

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import matplotlib.pyplot as plt
measures = [
    "Euclidean",
    "Manhattan",
    "Minkowski (p=3)",
    "Hamming"]
predictions = ["Buy", "Not Buy", "Buy", "Buy"]
values = [1, 0, 1, 1]
plt.figure(figsize=(9, 5))
```

```

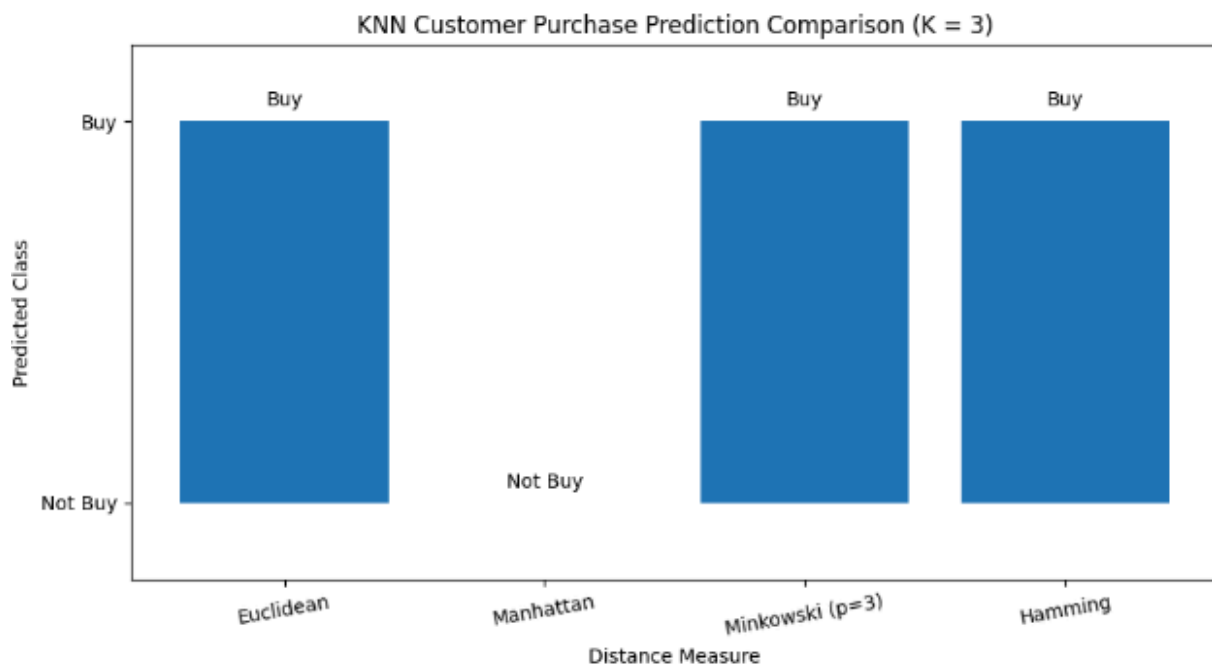
bars = plt.bar(measures, values)
plt.xlabel("Distance Measure")
plt.ylabel("Predicted Class")
plt.title("KNN Customer Purchase Prediction Comparison (K = 3)")
plt.yticks([0, 1], ["Not Buy", "Buy"])
plt.ylim(-0.2, 1.2)

```

```

for bar, label in zip(bars, predictions):
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 0.03,
        label,
        ha="center" )
plt.tight_layout()
plt.show()

```



Question 3: Medical Risk Classification

A hospital wants to classify whether a patient has High Risk or Low Risk based on medical measurements and symptoms.

Patient	Age	Blood Sugar Level	Fever	Cough	Fatigue	Class
P1	25	90	0	0	0	Low Risk
P2	30	95	0	1	0	Low Risk
P3	50	160	1	1	1	High Risk
P4	55	170	1	1	1	High Risk

Patient	Age	Blood Sugar Level	Fever	Cough	Fatigue	Class
P5	40	120	0	1	1	Low Risk
P6	60	180	1	1	1	High Risk

A new patient has:

Age	Blood Sugar Level	Fever	Cough	Fatigue
45	150	1	1	0

Using KNN with $K = 3$, classify the patient as High Risk or Low Risk using Euclidean, Manhattan, Minkowski, and Hamming distance.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

- Age
- Blood Sugar Level

Numerical = Age, Blood Sugar Level

Binary Features

- Fever
- Cough
- Fatigue

Binary = Fever, Cough, Fatigue

Target Variable

Class = High Risk / Low Risk

New patient:

(45,150,1,1,0)

b) Calculate the **Euclidean distance** between the new sample and all training samples

3) b) Euclidean distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$P1: \sqrt{(45-25)^2 + (150-90)^2 + (1-0)^2 + (1-0)^2 + (0-0)^2}$$

$$= 63.261$$

$$P2: \sqrt{(45-30)^2 + (150-95)^2 + (1-0)^2 + (1-1)^2 + (0-1)^2}$$

$$= 57.026$$

$$P3: \sqrt{(45-50)^2 + (150-160)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= 11.225$$

$$P4: \sqrt{(45-55)^2 + (150-170)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= 22.383$$

$$P5: \sqrt{(45-40)^2 + (150-120)^2 + (1-0)^2 + (1-1)^2 + (0-1)^2}$$

$$= 30.447$$

$$P6: \sqrt{(45-60)^2 + (150-180)^2 + (1-1)^2 + (1-1)^2 + (0-1)^2}$$

$$= 33.556$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(1) Manhattan Distance

$$d(x, y) = \sum |x_i - y_i|$$

$$p_1: 20 + 60 + 1 + 1 + 0 = 82$$

$$p_2: 15 + 55 + 1 + 0 + 0 = 71$$

$$p_3: 5 + 10 + 0 + 0 + 1 = 16$$

$$p_4: 10 + 20 + 0 + 0 + 1 = 31$$

$$p_5: 5 + 30 + 1 + 0 + 1 = 37$$

$$p_6: 15 + 30 + 0 + 0 + 1 = 46$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) Minkowski distance with $p=3$

formula:

$$d(x, y) = (\sum |x_i - y_i|^3)^{1/3}$$

$P_1: [20^3 + 60^3 + 1^3 + 1^3 + 0^3]^{1/3} = 60.735$
 $P_2: [15^3 + 55^3 + 1^3]^{1/3} = 55.349$
 $P_3: [5^3 + 10^3 + 1^3]^{1/3} = 10.406$
 $P_4: [10^3 + 20^3 + 1^3]^{1/3} = 20.805$
 $P_5: [5^3 + 30^3 + 1^3 + 1^3]^{1/3} = 30.045$
 $P_6: [15^3 + 30^3 + 1^3]^{1/3} = 31.215$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

Only binary features are used:

- Fever
- Cough
- Fatigue

New patient:

$X=(1,1,0)$

Formula

dH = Number of different binary values

Patient	Binary Features	Hamming Distance	Class
P1	(0,0,0)	2	Low Risk

P2	(0,1,0)	1	Low Risk
P3	(1,1,1)	1	High Risk
P4	(1,1,1)	1	High Risk
P5	(0,1,1)	2	Low Risk
P6	(1,1,1)	1	High Risk

There is a tie at distance 1.

Using dataset order:

P2, P3, P4

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	P3 (11.225)	P4 (22.383)	P5 (30.447)
Manhattan	P3 (16)	P4 (31)	P5 (37)
Minkowski p=3	P3 (10.406)	P4 (20.805)	P5 (30.045)
Hamming	P2 (1)	P3 (1)	P4 (1)

g) Predict the class label using majority voting for each distance measure.

Euclidean

- P3 → High Risk
- P4 → High Risk
- P5 → Low Risk

High=2, Low=1 Euclidean = HIGH RISK

Manhattan

- P3 → High Risk
- P4 → High Risk
- P5 → Low Risk

Manhattan = HIGH RISK

Minkowski p=3

- P3 → High Risk
- P4 → High Risk
- P5 → Low Risk

Minkowski = HIGH RISK

Hamming

- P2 → Low Risk
- P3 → High Risk
- P4 → High Risk

High=2,Low=1 Hamming = HIGH RISK

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import matplotlib.pyplot as plt
```

```
distance_measures = [
```

```
    "Euclidean",
```

```
    "Manhattan",
```

```
    "Minkowski (p=3)",
```

```
    "Hamming"]
```

```
prediction_score = [1, 1, 1, 1]
```

```
plt.figure(figsize=(10, 5))
```

```
plt.plot(
```

```
    distance_measures,
```

```
    prediction_score,
```

```
    marker='o',
```

```
    linewidth=2)
```

```
plt.xlabel("Distance Measure")
```

```
plt.ylabel("Risk Prediction")
```

```
plt.title("Comparison of KNN Distance Measures")
```

```
plt.yticks(
```

```
    [0, 1],
```

```
    ["Low Risk", "High Risk"])
```

```
plt.ylim(-0.1, 1.1)
```

```
for i, score in enumerate(prediction_score):
```

```
    plt.text(
```

```
        i,
```

```
        score + 0.03,
```

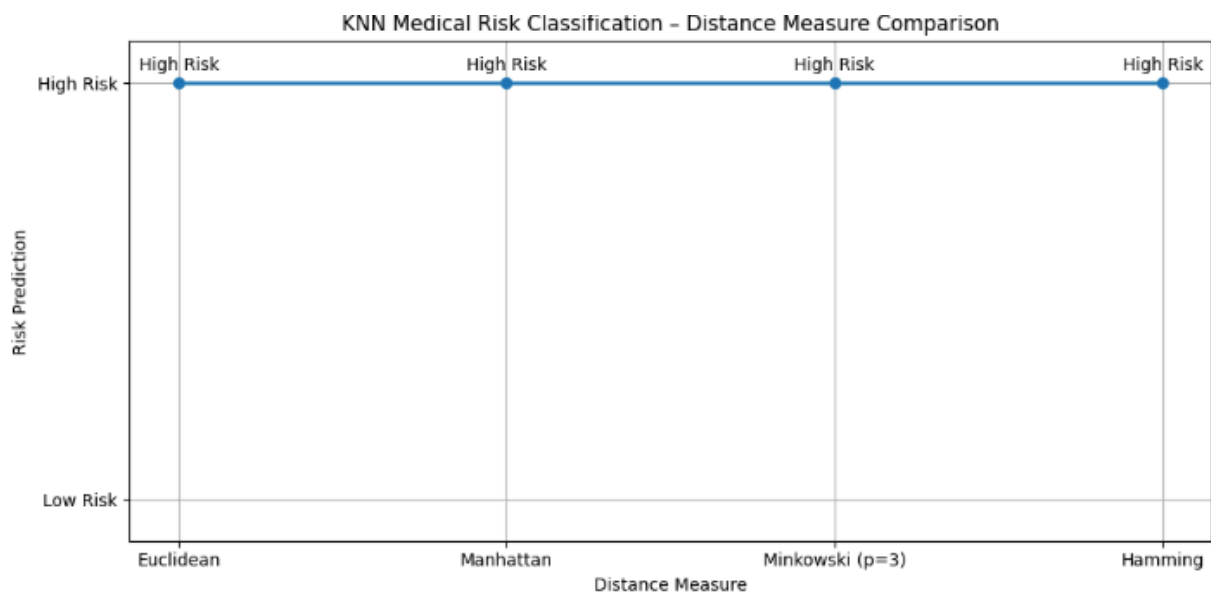
```
        "High Risk",
```

```
        ha="center" )
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

```
plt.show()
```



Question 4: Fruit Classification

A fruit shop wants to classify a fruit as Apple or Orange based on physical and binary characteristics.

Fruit	Weight	Diameter	Red Color	Citrus Smell	Smooth Skin	Class
F1	150	7	1	0	1	Apple
F2	160	8	1	0	1	Apple

Fruit	Weight	Diameter	Red Color	Citrus Smell	Smooth Skin	Class
F3	180	9	0	1	0	Orange
F4	200	10	0	1	0	Orange
F5	155	7.5	1	0	1	Apple
F6	190	9.5	0	1	0	Orange

A new fruit has:

Weight	Diameter	Red Color	Citrus Smell	Smooth Skin
170	8.5	1	0	1

Using KNN with $K = 3$, classify the fruit as Apple or Orange using all four distance measures.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

- **Weight**
- **Diameter**

Numerical = Weight, Diameter

Binary Features

- **Red Color**
- **Citrus Smell**
- **Smooth Skin**

Binary = Red Color, Citrus Smell, Smooth Skin

Target Variable

Class = Apple / Orange

b) Calculate the **Euclidean distance** between the new sample and all training samples

4) b) Euclidean Distance

Formula $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$

$$F1: \sqrt{(170 - 150)^2 + (8.5 - 7)^2 + (1 - 1)^2 + (0 - 0)^2 + (1 - 1)^2} = 20.056$$

$$F2: \sqrt{(170 - 160)^2 + (8.5 - 8)^2} = 10.012$$

$$F3: \sqrt{(170 - 180)^2 + (8.5 - 9)^2 + (1 - 0)^2 + (0 - 1)^2 + (1 - 0)^2} = 10.161$$

$$F4: \sqrt{(170 - 200)^2 + (8.5 - 10)^2 + 1^2 + 1^2 + 1^2} = 30.087$$

$$F5: \sqrt{(170 - 155)^2 + (8.5 - 7.5)^2} = 15.033$$

$$F6: \sqrt{(170 - 190)^2 + (8.5 - 9.5)^2 + 1^2 + 1^2 + 1^2} = 20.100$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

• (c) Manhattan Distance

$$d(x, y) = \sum |x_i - y_i|$$

$$F1: 20 + 1.5 + 0 = 21.5$$

$$F2: 10 + 0.5 + 0 = 10.5$$

$$F3: 10 + 0.5 + 1 + 1 + 1 = 13.5$$

$$F4: 30 + 1.5 + 1 + 1 + 1 = 34.5$$

$$F5: 15 + 1 + 0 + 0 = 16$$

$$2 F6: 20 + 1 + 1 + 1 + 1 = 24$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) Minkowski Distance ($p=3$)

$$d(x, y) = (\sum |x_i - y_i|^p)^{1/p}$$

$$F1: [(20)^3 + (1.5)^3]^{1/3}$$

$$= \sqrt[3]{8003.375}$$

$$= 20.003$$

$$F2: [(10)^3 + (0.5)^3]^{1/3} = 10.0$$

$$F3: [(10)^3 + (0.5)^3 + 1^3 + 1^3 + 1^3]^{1/3}$$

$$= 10.010$$

$$F4: [(30)^3 + (1.5)^3 + 1^3 + 1^3 + 1^3]^{1/3}$$

$$= 30.002$$

$$F5: [(15)^3 + (1)^3]^{1/3} = 15.001$$

$$F6: [(20)^3 + (1)^3 + 1^3 + 1^3 + 1^3]^{1/3}$$

$$= \sqrt[3]{8003}$$

$$= 20.003$$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only:

- Red Color
- Citrus Smell
- Smooth Skin

New fruit:

$X=(1,0,1)$

Calculations

Fruit	Binary Features	Hamming Distance	Class
F1	(1,0,1)	0	Apple
F2	(1,0,1)	0	Apple
F3	(0,1,0)	3	Orange
F4	(0,1,0)	3	Orange
F5	(1,0,1)	0	Apple
F6	(0,1,0)	3	Orange

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	F2 (10.012)	F3 (10.161)	F5 (15.033)
Manhattan	F2 (10.5)	F3 (13.5)	F5 (16)
Minkowski (p=3)	F2 (10.000)	F3 (10.010)	F5 (15.001)
Hamming	F1 (0)	F2 (0)	F5 (0)

g) Predict the class label using majority voting for each distance measure.

Euclidean

- F2 → Apple
- F3 → Orange
- F5 → Apple

Apple=2, Orange=1 Euclidean = APPLE

Manhattan

- F2 → Apple
- F3 → Orange
- F5 → Apple

Manhattan = APPLE

Minkowski

- $F2 \rightarrow \text{Apple}$
- $F3 \rightarrow \text{Orange}$
- $F5 \rightarrow \text{Apple}$

Minkowski = APPLE

Hamming

- $F1 \rightarrow \text{Apple}$
- $F2 \rightarrow \text{Apple}$
- $F5 \rightarrow \text{Apple}$

Apple=3, Orange=0 Hamming = APPLE

h) Compare the results and explain which distance measure is most suitable for the given scenario.

import matplotlib.pyplot as plt

```
distance_measures = [
    "Euclidean",
    "Manhattan",
    "Minkowski (p=3)",
    "Hamming"
]
```

```
prediction_values = [1, 1, 1, 1]
predictions = ["Apple", "Apple", "Apple", "Apple"]
```

```
plt.figure(figsize=(10, 5))
```

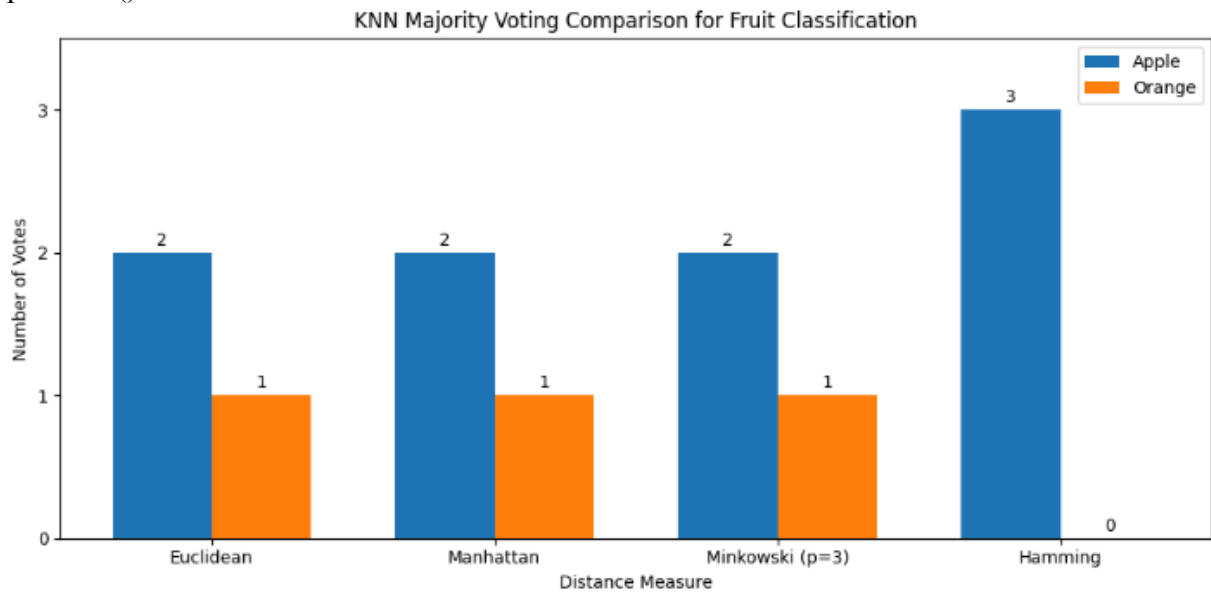
```
plt.plot(
    distance_measures,
    prediction_values,
    marker="o",
    linewidth=2
)
```

```
plt.xlabel("Distance Measure")
plt.ylabel("Predicted Class")
plt.title("KNN Fruit Classification Comparison (K = 3)")
```

```
plt.yticks([0, 1], ["Orange", "Apple"])
plt.ylim(-0.1, 1.1)
```

```
for i, label in enumerate(predictions):
    plt.text(
        i,
        prediction_values[i] + 0.03,
        label,
        ha="center"
    )
```

```
plt.grid(True)
plt.tight_layout()
plt.show()
```



Question 5: House Price Category Prediction

A real estate company wants to classify houses as Low Price or High Price.

House	Area in sq.ft	Bedrooms	Parking Available	Near Main Road	Garden Available	Category
H1	800	2	0	0	0	Low Price
H2	1000	2	1	0	0	Low Price
H3	1500	3	1	1	1	High Price
H4	1800	4	1	1	1	High Price
H5	1200	3	0	1	0	Low Price
H6	2000	4	1	1	1	High Price

A new house has:

Area	Bedrooms	Parking Available	Near Main Road	Garden Available
1400	3	1	1	0

Using KNN with $K = 3$, classify the house as Low Price or High Price using Euclidean, Manhattan, Minkowski, and Hamming distance.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

- Area in sq.ft

- **Bedrooms**

Numerical = Area, Bedrooms

Binary Features

- **Parking Available**
- **Near Main Road**
- **Garden Available**

Binary = Parking, Main Road, Garden

Target Variable

Category = Low Price / High Price

b) Calculate the **Euclidean distance** between the new sample and all training samples

5) b) Euclidean distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$= \sqrt{(1400 - 800)^2 + (3 - 2)^2 + (1 - 0)^2 + (1 - 0)^2 + (0 - 0)^2} = 600.00$$

$$= \sqrt{(1400 - 1000)^2 + (3 - 2)^2 + (1 - 1)^2 + (1 - 0)^2} = 400.002$$

$$= \sqrt{(1400 - 1500)^2 + (3 - 3)^2 + (1 - 1)^2 + (1 - 1)^2 + (0 - 1)^2}$$

$$= 100.005$$

$$= \sqrt{(1400 - 1800)^2 + (3 - 4)^2 + (1 - 1)^2 + (1 - 1)^2 + (0 - 1)^2}$$

$$= 400.002$$

$$= \sqrt{(1400 - 1200)^2 + (3 - 3)^2 + (1 - 0)^2 + (1 - 1)^2 + (0 - 0)^2}$$

$$= 200.002$$

$$= \sqrt{(1400 - 2000)^2 + (3 - 4)^2 + (1 - 1)^2 + (1 - 1)^2 + (0 - 1)^2}$$

$$= 600.002$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

c) Manhattan distance

$$d(x, y) = \sum |x_i - y_i|$$

$$H_1: 600 + 1 + 1 + 1 = 603$$

$$H_2: 400 + 1 + 1 = 402$$

$$H_3: 100 + 1 = 101$$

$$H_4: 400 + 1 + 1 = 402$$

$$H_5: 200 + 1 = 201$$

$$H_6: 600 + 1 + 1 = 602$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) minkowski distance with $p=3$

$$d(x, y) = \left[\sum |x_i - y_i|^3 \right]^{1/3}$$

(1) $[600^3 + 1^3 + 1^3 + 1^3]^{1/3} = 600.000$

(2) $[400^3 + 1^3 + 0^3 + 1^3]^{1/3} = 400.000$

(3) $[100^3 + 0 + 0 + 0 + 1^3]^{1/3} = 10.000$

(4) $[400^3 + 1^3 + 0 + 0 + 1^3]^{1/3} = 400.000$

(5) $[200^3 + 0 + 1^3 + 0 + 0]^{1/3} = 200.000$

(6) $[600^3 + 1^3 + 0 + 0 + 1^3]^{1/3} = 600.000$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only binary features:

- Parking Available
- Near Main Road
- Garden Available

New house:

$X=(1,1,0)$

Hamming Formula

dH =Number of different binary positions

House	Binary Features	Hamming Distance	Category
H1	(0,0,0)	2	Low Price
H2	(1,0,0)	1	Low Price
H3	(1,1,1)	1	High Price
H4	(1,1,1)	1	High Price
H5	(0,1,0)	1	Low Price
H6	(1,1,1)	1	High Price

There is a tie among five houses at distance 1. Using dataset order, select:

H2, H3, H4

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	H3 (100.005)	H5 (200.002)	H2/H4 (400.002)
Manhattan	H3 (101)	H5 (201)	H2/H4 (402)
Minkowski p=3	H3 (100.000)	H5 (200.000)	H2/H4 (400.000)
Hamming	H2 (1)	H3 (1)	H4 (1)

For the tie, H2 is selected before H4.

g) Predict the class label using majority voting for each distance measure.

Euclidean

- H3 → High Price
- H5 → Low Price
- H2 → Low Price

High=1, Low=2 Euclidean = LOW PRICE

Manhattan

- H3 → High Price
- H5 → Low Price
- H2 → Low Price

Manhattan = LOW PRICE

Minkowski

- H3 → High Price
- H5 → Low Price
- H2 → Low Price

Minkowski = LOW PRICE

Hamming

- H2 → Low Price
- H3 → High Price
- H4 → High Price

High=2,Low=1 Hamming = HIGH PRICE

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
distance_measures = [
```

```
    "Euclidean",
```

```
    "Manhattan",
```

```
    "Minkowski (p=3)",
```

```
    "Hamming"
```

```
]
```

```
low_price_votes = [2, 2, 2, 1]
```

```
high_price_votes = [1, 1, 1, 2]
```

```
x = np.arange(len(distance_measures))
```

```
width = 0.35
```

```
plt.figure(figsize=(10, 5))
```

```
plt.bar(
```

```
    x - width/2,
```

```
    low_price_votes,
```

```
    width,
```

```
    label="Low Price")
```

```
plt.bar(
```

```
    x + width/2,
```

```
    high_price_votes,
```

```
    width,
```

```
    label="High Price")
```

```
plt.xlabel("Distance Measure")

plt.ylabel("Number of Votes")

plt.title("KNN House Price Classification - Majority Voting")

plt.xticks(x, distance_measures)

plt.yticks([0, 1, 2, 3])

plt.ylim(0, 3.5)

for i, value in enumerate(low_price_votes):

    plt.text(

        i - width/2,

        value + 0.05,

        str(value),

        ha="center")

for i, value in enumerate(high_price_votes):

    plt.text(

        i + width/2,

        value + 0.05,

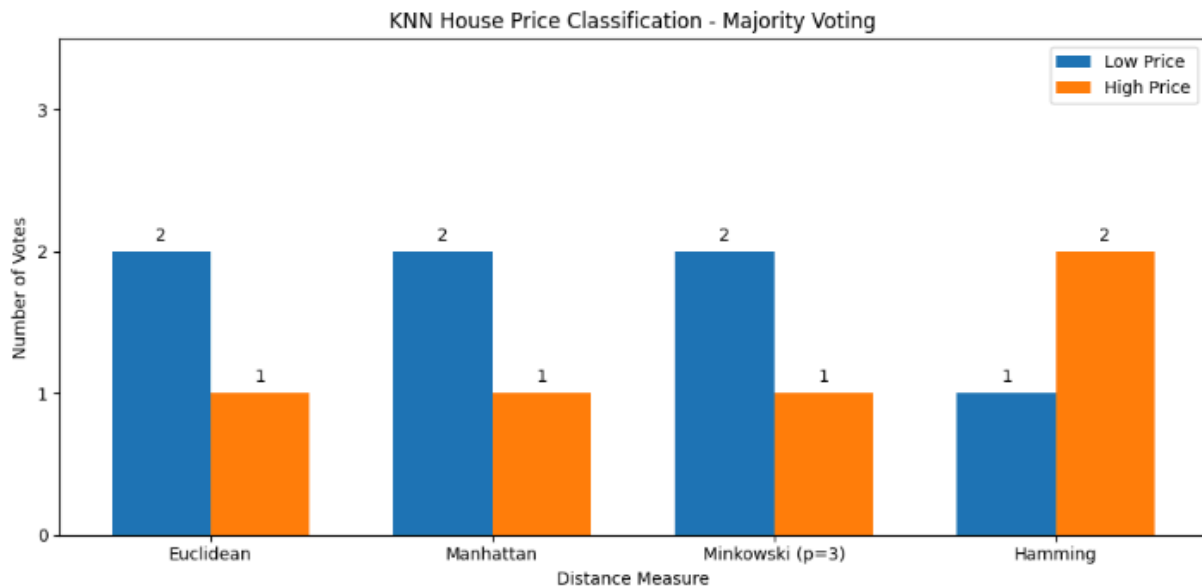
        str(value),

        ha="center")

plt.legend()

plt.tight_layout()

plt.show()
```



Question 6: Loan Approval Prediction

A bank wants to predict whether a loan application should be Approved or Rejected.

Applicant	Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default	Class
A1	25000	580	0	1	1	Rejected
A2	30000	600	0	1	0	Rejected
A3	60000	750	1	0	0	Approved
A4	70000	800	1	0	0	Approved
A5	40000	650	0	1	0	Rejected
A6	80000	820	1	0	0	Approved

A new applicant has:

Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default
55000	700	1	0	0

Using KNN with $K = 3$, classify the loan application as Approved or Rejected using all four distance measures.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

1. Monthly Income
2. Credit Score

Numerical Features = Monthly Income, Credit Score

Binary Features

1. Govt Job
2. Existing Loan
3. Past Default

Binary Features = Govt Job, Existing Loan, Past Default

Target Variable

Class = Approved / Rejected

b) Calculate the **Euclidean distance** between the new sample and all training samples

b) b) Euclidean Distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$= \sqrt{(55000 - 25000)^2 + (700 - 500)^2 + (1 - 0)^2 + (0 - 12)^2 + (0 - 1)^2}$$

$$= 30000.240$$

$$= \sqrt{25000^2 + 100^2 + 1^2 + 1^2}$$

$$= 25000.200$$

$$= \sqrt{5000^2 + 50^2}$$

$$= 5000.250$$

$$= \sqrt{15000^2 + 100^2}$$

$$= 15000.333$$

$$= \sqrt{15000^2 + 50^2 + 1^2 + 1^2}$$

$$= 15000.083$$

$$= \sqrt{25000^2 + 120^2}$$

$$= 25000.288$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(1) Manhattan distance

$$d(x, y) = \sum |x_i - y_i|$$

$$(1) : 3000 + 120 + 1 + 1 + 1 = 30123$$

$$(2) : 25000 + 100 + 1 + 1 = 25102$$

$$(3) : 5000 + 50 = 5050$$

$$(4) : 15000 + 100 = 15100$$

$$(5) : 15000 + 50 + 1 + 1 = 15052$$

$$(6) : 25000 + 120 = 25120$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

d) Minkowski distance with $p=3$

$$d(x, y) = \left(\sum (x_i - y_i)^3 \right)^{1/3}$$

$$c1: [(30000)^3 + (120)^3 + 1^3 + 1^3 + 1^3]^{1/3} \\ = 30000.001$$

$$c2: [(25000)^3 + (100)^3 + 1^3 + 1^3]^{1/3} \\ = 25000.001$$

$$c3: [(5000)^3 + (50)^3]^{1/3} \\ = 5000.002$$

$$c4: [(15000)^3 + (100)^3]^{1/3} \\ = 15000.001$$

$$c5: [(13000)^3 + (50)^3 + 1^3 + 1^3]^{1/3} \\ = 13000.000$$

$$c6: [(25000)^3 + (120)^3]^{1/3} \\ = 25000.001$$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only the binary features:

- Govt Job
- Existing Loan

- Past Default

New applicant:

$X=(1,0,0)$

Calculations

Applicant	Binary Features	Hamming Distance	Class
A1	(0,1,1)	3	Rejected
A2	(0,1,0)	2	Rejected
A3	(1,0,0)	0	Approved
A4	(1,0,0)	0	Approved
A5	(0,1,0)	2	Rejected
A6	(1,0,0)	0	Approved

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	A3 (5000.250)	A5 (15000.083)	A4 (15000.333)
Manhattan	A3 (5050)	A5 (15052)	A4 (15100)
Minkowski p=3	A3 (5000.002)	A5 (15000.000)	A4 (15000.001)
Hamming	A3 (0)	A4 (0)	A6 (0)

g) Predict the class label using majority voting for each distance measure.

Euclidean

- A3 → Approved
- A5 → Rejected
- A4 → Approved

Approved=2, Rejected=1 Euclidean = APPROVED

Manhattan

- A3 → Approved
- A5 → Rejected
- A4 → Approved

Manhattan = APPROVED

Minkowski

- A3 → Approved
- A5 → Rejected
- A4 → Approved

Minkowski = APPROVED

Hamming

- A3 → Approved
- A4 → Approved
- A6 → Approved

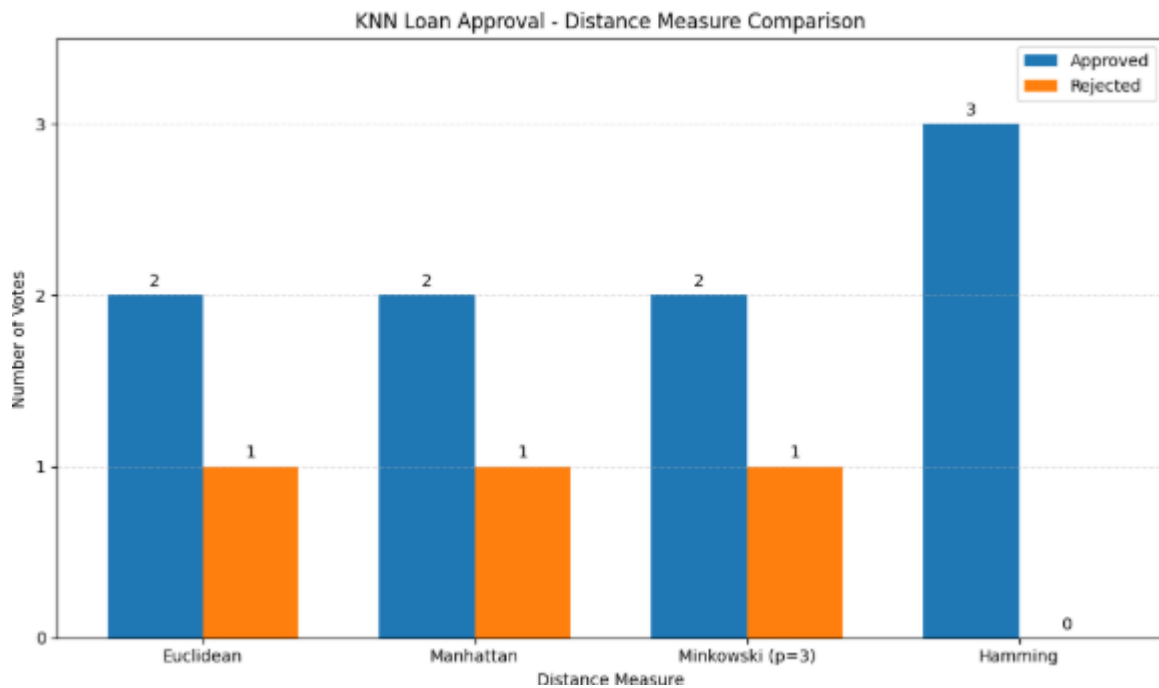
Approved=3, Rejected=0 Hamming = APPROVED

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import matplotlib.pyplot as plt
import numpy as np
distance_measures=["Euclidean","Manhattan","Minkowski (p=3)","Hamming"]
approved_votes=[2,2,2,3]
rejected_votes=[1,1,1,0]
x=np.arange(len(distance_measures))
width=0.35
```

```
plt.figure(figsize=(10,6))
bars1=plt.bar(x-width/2,approved_votes,width,label="Approved")
bars2=plt.bar(x+width/2,rejected_votes,width,label="Rejected")
plt.title("KNN Loan Approval - Distance Measure Comparison")
plt.xlabel("Distance Measure")
plt.ylabel("Number of Votes")
plt.xticks(x,distance_measures)
plt.yticks([0,1,2,3])
plt.ylim(0,3.5)

for bar in bars1:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
for bar in bars2:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
plt.legend()
plt.grid(axis="y",linestyle="--",alpha=0.4)
plt.tight_layout()
plt.show()
```



Question 7: Movie Preference Classification

A movie platform wants to classify a user as an Action Lover or Comedy Lover.

User	Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom	Class
U1	5	1	1	0	0	Action Lover
U2	4	2	1	0	0	Action Lover
U3	1	5	0	1	1	Comedy Lover
U4	2	4	0	1	1	Comedy Lover
U5	5	2	1	0	0	Action Lover
U6	1	4	0	1	1	Comedy Lover

A new user has:

Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom
3	3	1	1	0

Using KNN with $K = 3$, classify the new user as Action Lover or Comedy Lover using Euclidean, Manhattan, Minkowski, and Hamming distance.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

1. Action Rating
2. Comedy Rating

Numerical Features = Action Rating, Comedy Rating

Binary Features

1. Watched Superhero Movie
2. Watched Stand-up Show
3. Watched Sitcom

Binary Features = Superhero, Stand-up, Sitcom

Target Variable

Class = Action Lover / Comedy Lover

b) Calculate the **Euclidean distance** between the new sample and all training samples

7) b) Euclidean Distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$\sqrt{(3-5)^2 + (3-1)^2 + (1-1)^2 + (2-0)^2 + (0-0)^2}$$

$$d(x, u_1) = 3$$

$$\sqrt{(3-4)^2 + (3-2)^2 + (1-1)^2 + (1-0)^2 + (0-0)^2}$$

$$d(x, u_2) = 1.732$$

$$\sqrt{(3-1)^2 + (3-5)^2 + (1-0)^2 + (1-1)^2 + (0-1)^2}$$

$$d(x, u_3) = 3.162$$

$$\sqrt{(3-2)^2 + (3-4)^2 + (1-0)^2 + (1-1)^2 + (0-1)^2}$$

$$d(x, u_4) = 2$$

$$\sqrt{(3-5)^2 + (3-2)^2 + (1-1)^2 + (1-0)^2}$$

$$d(x, u_5) = 2.449$$

$$\sqrt{(3-1)^2 + (3-4)^2 + (1-0)^2 + (1-1)^2}$$

$$d(x, u_6) = 2.646$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(c) manhattan Distance

$$d(x, y) = \sum |x_i - y_i|$$

U1: $2 + 2 + 1 = 6$

U2: $1 + 1 + 1 = 3$

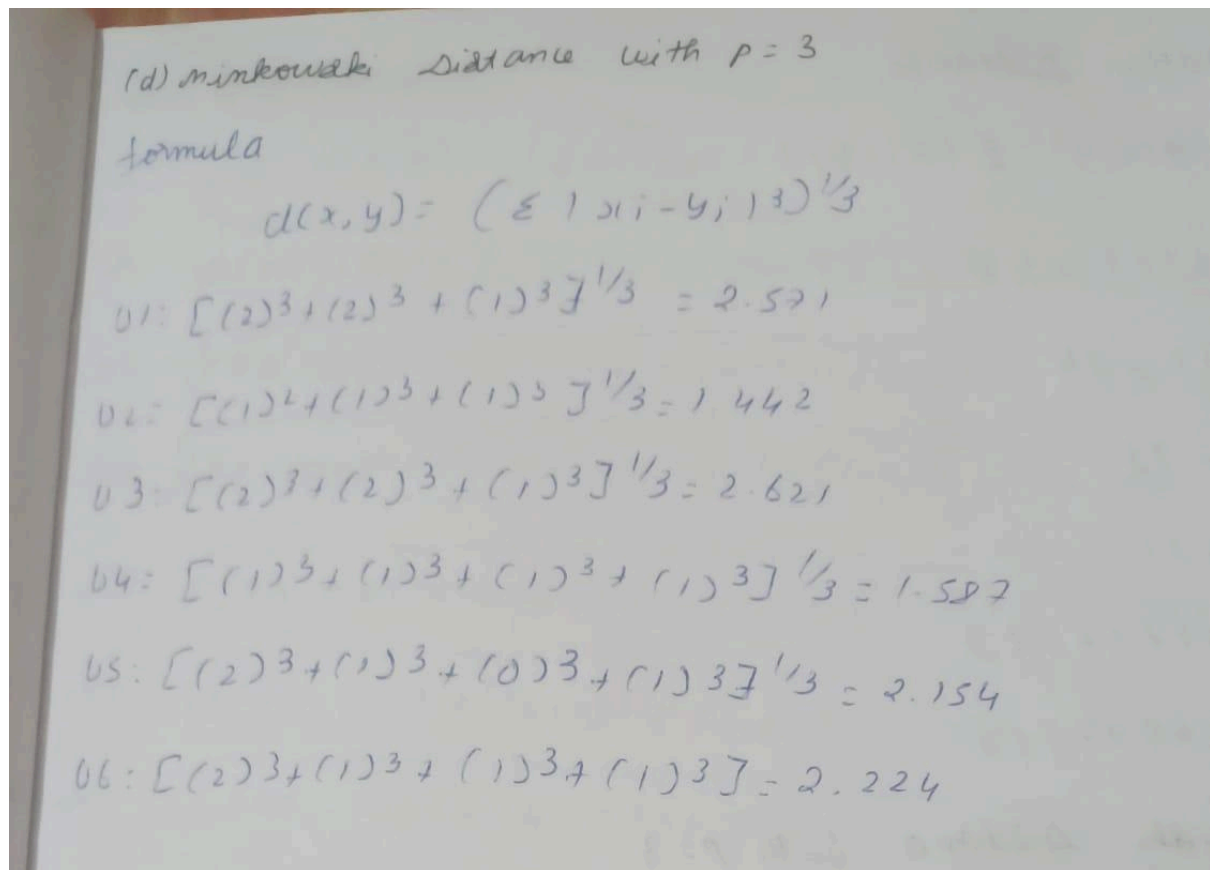
U3: $2 + 2 + 1 + 1 = 6$

U4: $1 + 1 + 1 + 1 = 4$

U5: $2 + 1 + 1 = 4$

U6: $2 + 1 + 1 + 1 = 5$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.



e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only binary features:

- Watched Superhero Movie
- Watched Stand-up Show
- Watched Sitcom

New user:

$X=(1,1,0)$

Formula

dH=Number of different binary features

User	Binary Features	Hamming Distance	Class
U1	(1,0,0)	1	Action Lover
U2	(1,0,0)	1	Action Lover
U3	(0,1,1)	2	Comedy Lover
U4	(0,1,1)	2	Comedy Lover
U5	(1,0,0)	1	Action Lover

U6 (0,1,1) 2 Comedy Lover

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	U2 (1.732)	U4 (2.000)	U5 (2.449)
Manhattan	U2 (3)	U4 (4)	U5 (4)
Minkowski (p=3)	U2 (1.442)	U4 (1.587)	U5 (2.154)
Hamming	U1 (1)	U2 (1)	U5 (1)

g) Predict the class label using majority voting for each distance measure.

Euclidean

- U2 → Action Lover
- U4 → Comedy Lover
- U5 → Action Lover

Action=2,Comedy=1 Euclidean = ACTION LOVER

Manhattan

- U2 → Action Lover
- U4 → Comedy Lover
- U5 → Action Lover

Action=2,Comedy=1 Manhattan = ACTION LOVER

Minkowski p=3

- U2 → Action Lover
- U4 → Comedy Lover
- U5 → Action Lover

Action=2,Comedy=1 Minkowski = ACTION LOVER

Hamming

- U1 → Action Lover
- U2 → Action Lover
- U5 → Action Lover

Action=3,Comedy=0 Hamming = ACTION LOVER

h) Compare the results and explain which distance measure is most suitable for the given scenario.

import matplotlib.pyplot as plt

```

import numpy as np
distance_measures=["Euclidean","Manhattan","Minkowski (p=3)","Hamming"]
approved_votes=[2,2,2,3]
rejected_votes=[1,1,1,0]
x=np.arange(len(distance_measures))
width=0.35

```

```

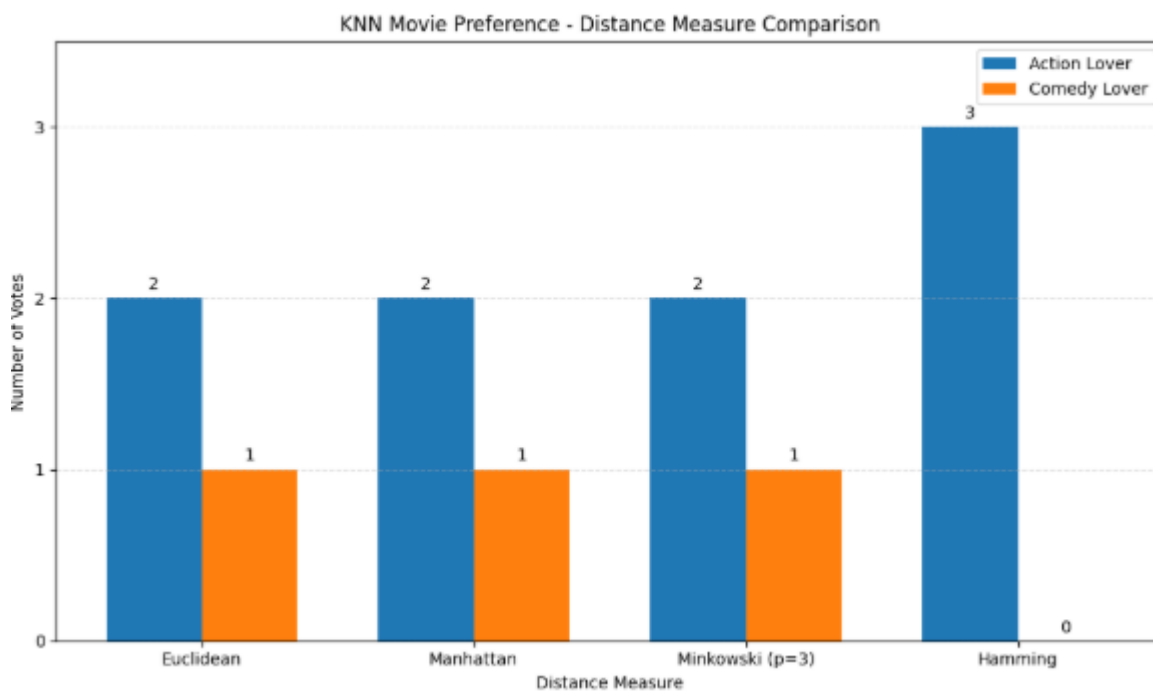
plt.figure(figsize=(10,6))
bars1=plt.bar(x-width/2,approved_votes,width,label="Approved")
bars2=plt.bar(x+width/2,rejected_votes,width,label="Rejected")
plt.title("KNN Loan Approval - Distance Measure Comparison")
plt.xlabel("Distance Measure")
plt.ylabel("Number of Votes")
plt.xticks(x,distance_measures)
plt.yticks([0,1,2,3])
plt.ylim(0,3.5)

```

```

for bar in bars1:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
for bar in bars2:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
plt.legend()
plt.grid(axis="y",linestyle="--",alpha=0.4)
plt.tight_layout()
plt.show()

```



Question 8: Crop Disease Detection

An agriculture research center wants to classify a plant as Healthy or Diseased.

Plant	Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting	Class
L1	2	80	0	0	0	Healthy
L2	3	75	0	0	0	Healthy
L3	12	40	1	1	1	Diseased
L4	15	35	1	1	1	Diseased
L5	5	70	0	0	1	Healthy
L6	14	38	1	1	1	Diseased

A new plant has:

Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting
10	50	1	1	0

Using KNN with $K = 3$, classify the plant as Healthy or Diseased using all four distance measures.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

1. Leaf Spot Count
2. Leaf Color Index

Numerical Features = Leaf Spot Count, Leaf Color Index

Binary Features

1. Yellow Leaves
2. White Fungus
3. Wilting

Binary Features = Yellow Leaves, White Fungus, Wilting

Target Variable

Class = Healthy / Diseased

b) Calculate the **Euclidean distance** between the new sample and all training samples

8) b) Euclidean Distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$L1 = \sqrt{(10-2)^2 + (50-80)^2 + (1-0)^2 + (1-0)^2 + (-0-0)^2} \\ = 31.081$$

$$L2 = \sqrt{(10-3)^2 + (50-75)^2 + (1-0)^2 + (1-0)^2} = 26.000$$

$$L3 = \sqrt{(10-12)^2 + (50-40)^2 + (1-1)^2 + (1-1)^2} = 10.247$$

$$L4 = \sqrt{(10-15)^2 + (50-35)^2 + (1-1)^2 + (1-1)^2} = 15.843$$

$$L5 = \sqrt{(10-5)^2 + (50-70)^2 + (1-0)^2 + (1-0)^2} \\ = 20.687$$

$$L6 = \sqrt{(10-14)^2 + (50-38)^2 + (1-1)^2 + (1-1)^2} \\ = 12.689$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(c) Manhattan Distance

$$d(x, y) = \sum |x_i - y_i|$$

$$8 + 30 + 1 + 1 = 40$$

$$7 + 25 + 1 + 1 = 34$$

$$2 + 10 + 1 = 13$$

$$5 + 15 + 1 = 21$$

$$5 + 20 + 1 + 1 = 27$$

$$4 + 12 + 0 + 0 + 1 = 17$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) minkowski distance with $p=3$

$$d(x,y) = (\sum |x_i - y_i|^3)^{1/3}$$

$$[(8)^3 + (30)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 30.189$$

$$[(7)^3 + (25)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 25.183$$

$$[(2)^3 + (10)^3 + (1)^3]^{1/3}$$

$$= 10.030$$

$$[(5)^3 + (15)^3 + (1)^3]^{1/3}$$

$$= 15.184$$

$$[(5)^3 + (20)^3 + (1)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 20.106$$

$$[(4)^3 + (12)^3 + (1)^3]^{1/3}$$

$$= 12.149$$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

Plant	Binary Features	Hamming Distance	Class
-------	-----------------	------------------	-------

L1	(0,0,0)	2	Healthy
L2	(0,0,0)	2	Healthy
L3	(1,1,1)	1	Diseased
L4	(1,1,1)	1	Diseased
L5	(0,0,1)	3	Healthy
L6	(1,1,1)	1	Diseased

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st Neighbour	2nd Neighbour	3rd Neighbour
Euclidean	L3 (10.247)	L6 (12.689)	L4 (15.843)
Manhattan	L3 (13)	L6 (17)	L4 (21)
Minkowski p=3	L3 (10.030)	L6 (12.149)	L4 (15.184)
Hamming	L3 (1)	L4 (1)	L6 (1)

g) Predict the class label using majority voting for each distance measure.

Euclidean

Nearest neighbours:

- L3 → Diseased
- L6 → Diseased
- L4 → Diseased

Diseased=3, Healthy=0 Euclidean = DISEASED

Manhattan

Nearest neighbours:

- L3 → Diseased
- L6 → Diseased
- L4 → Diseased

Manhattan = DISEASED

Minkowski p=3

Nearest neighbours:

- L3 → Diseased
- L6 → Diseased
- L4 → Diseased

Minkowski = DISEASED

Hamming

Nearest neighbours:

- L3 → Diseased
- L4 → Diseased
- L6 → Diseased

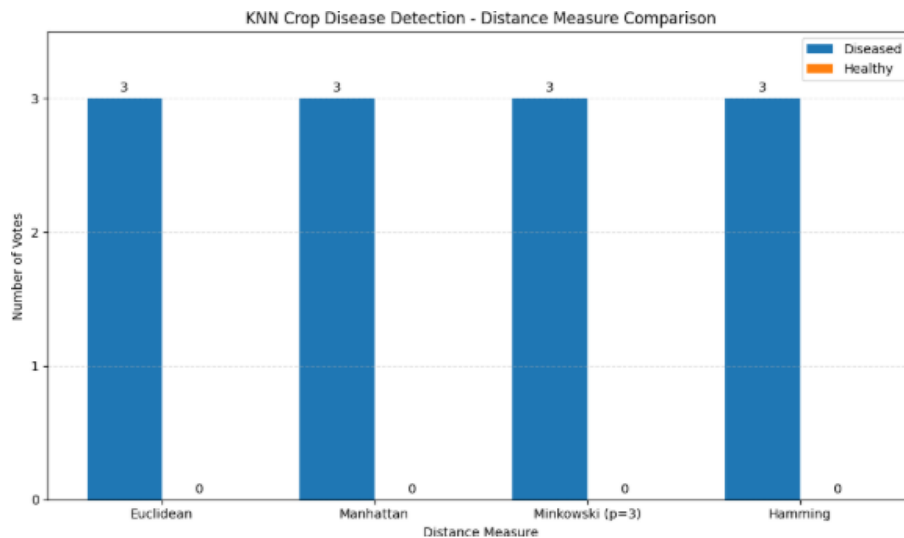
Diseased=3, Healthy=0 Hamming = DISEASED

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import numpy as np
distance_measures=["Euclidean","Manhattan","Minkowski (p=3)","Hamming"]
approved_votes=[2,2,2,3]
rejected_votes=[1,1,1,0]
x=np.arange(len(distance_measures))
width=0.35

plt.figure(figsize=(10,6))
bars1=plt.bar(x-width/2,approved_votes,width,label="Approved")
bars2=plt.bar(x+width/2,rejected_votes,width,label="Rejected")
plt.title("KNN Loan Approval - Distance Measure Comparison")
plt.xlabel("Distance Measure")
plt.ylabel("Number of Votes")
plt.xticks(x,distance_measures)
plt.yticks([0,1,2,3])
plt.ylim(0,3.5)

for bar in bars1:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
for bar in bars2:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
plt.legend()
plt.grid(axis="y",linestyle="--",alpha=0.4)
plt.tight_layout()
plt.show()
```



Question 9: Network Intrusion Detection

A cybersecurity system wants to classify network activity as Normal or Attack.

Record	Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP	Class
N1	20	0	0	0	0	Normal
N2	30	1	0	0	0	Normal
N3	100	5	1	1	1	Attack
N4	120	6	1	1	1	Attack
N5	40	1	0	0	0	Normal
N6	110	7	1	1	1	Attack

A new network record has:

Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP
90	4	1	1	0

Using KNN with $K = 3$, classify the network activity as Normal or Attack using Euclidean, Manhattan, Minkowski, and Hamming distance.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

1. Packets per Second
2. Failed Login Attempts

Numerical = Packets per Second, Failed Login Attempts

Binary Features

1. Unknown Port Used
2. Admin Access Tried
3. Blacklisted IP

Binary = Unknown Port, Admin Access, Blacklisted IP

Target Variable

Class = Normal / Attack

b) Calculate the **Euclidean distance** between the new sample and all training samples

q) b) Euclidean Distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$N1: \sqrt{(90-20)^2 + (4-0)^2 + (1-0)^2 + (1-0)^2} = 70.128$$

$$N2: \sqrt{(90-30)^2 + (4-1)^2 + (1-0)^2 + (1-0)^2} = 60.092$$

$$N3: \sqrt{(90-100)^2 + (4-5)^2 + (1-1)^2 + (1-1)^2} = 10.100$$

$$N4: \sqrt{(90-120)^2 + (4-6)^2 + (1-1)^2 + (1-1)^2} = 30.083$$

$$N5: \sqrt{(90-40)^2 + (4-1)^2 + (1-0)^2 + (1-0)^2} = 50.110$$

$$N6: \sqrt{(90-110)^2 + (4-7)^2 + (1-1)^2 + (1-1)^2} = 20.248$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

(1) manhattan distance

$$d(x, y) = \sum |x_i - y_i|$$

$$70 + 4 + 1 + 1 = 76$$

$$60 + 3 + 1 + 1 = 65$$

$$10 + 1 + 1 = 12$$

$$30 + 2 + 1 = 33$$

$$50 + 3 + 1 + 1 = 55$$

$$20 + 3 + 1 = 24$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

d) Minkowski Distance with $p=3$

$$d(x, y) = (\sum (x_i - y_i)^3)^{1/3}$$

$$w_1: [(70)^3 + (4)^3 + (1)^3 + (1)^3]^{1/3} = 70$$

$$w_2: [(60)^3 + (3)^3 + (1)^3 + (1)^3]^{1/3} = 60.002$$

$$N3: [(10)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 10.007$$

$$N4: [(30)^3 + (2)^3 + (1)^3]^{1/3}$$

$$= 30.003$$

$$N5: [(50)^3 + (3)^3 + (1)^3 + (1)^3]^{1/3}$$

$$= 50.004$$

$$N6: [(20)^3 + (3)^3 + (1)^3]^{1/3}$$

$$= 20.003$$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only the binary features:

- Unknown Port Used
- Admin Access Tried
- Blacklisted IP

New record:

$X=(1,1,0)$

Calculations

Record	Binary Features	Hamming Distance	Class
N1	(0,0,0)	2	Normal
N2	(0,0,0)	2	Normal
N3	(1,1,1)	1	Attack
N4	(1,1,1)	1	Attack
N5	(0,0,0)	2	Normal
N6	(1,1,1)	1	Attack

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure	1st	2nd	3rd
Euclidean	N3 (10.100)	N6 (20.248)	N4 (30.083)
Manhattan	N3 (12)	N6 (24)	N4 (33)
Minkowski p=3	N3 (10.007)	N6 (20.003)	N4 (30.003)
Hamming	N3 (1)	N4 (1)	N6 (1)

All selected nearest neighbours are **Attack** records.

g) Predict the class label using majority voting for each distance measure.

Euclidean

- N3 → Attack
- N6 → Attack
- N4 → Attack

Attack=3, Normal=0 Euclidean = ATTACK

Manhattan

- N3 → Attack
- N6 → Attack
- N4 → Attack

Manhattan = ATTACK

Minkowski

- N3 → Attack
- N6 → Attack
- N4 → Attack

Minkowski = ATTACK

Hamming

- N3 → Attack
- N4 → Attack
- N6 → Attack

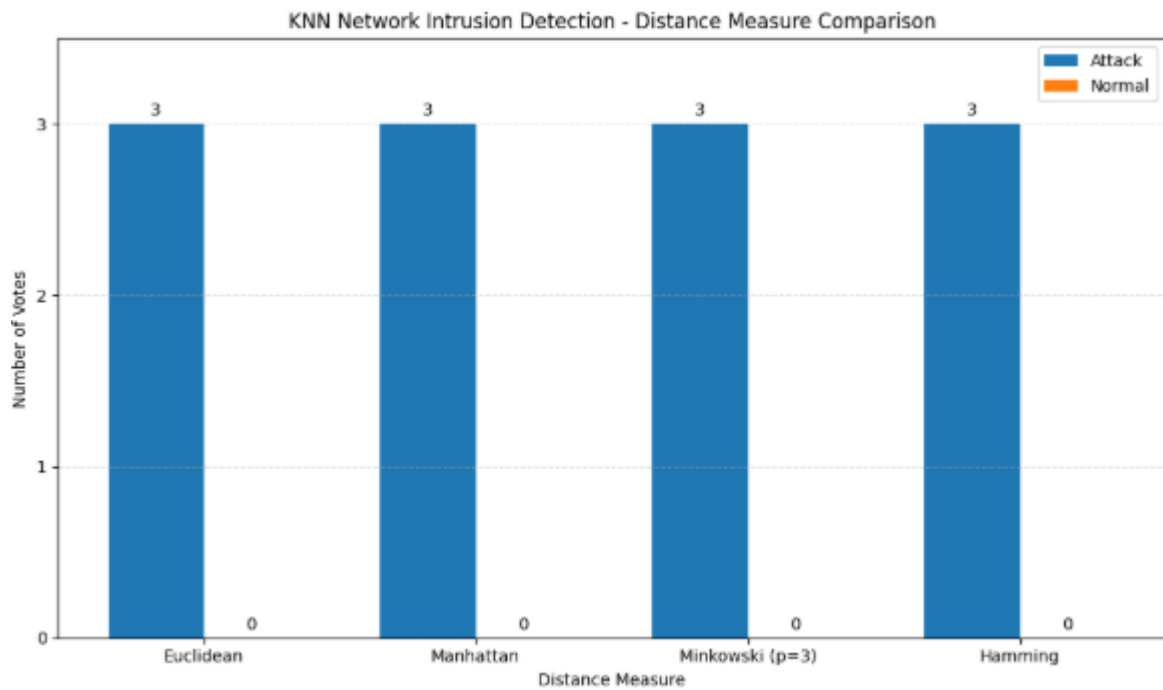
Attack=3,Normal=0 Hamming = ATTACK

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import numpy as np
distance_measures=["Euclidean","Manhattan","Minkowski (p=3)","Hamming"]
approved_votes=[2,2,2,3]
rejected_votes=[1,1,1,0]
x=np.arange(len(distance_measures))
width=0.35

plt.figure(figsize=(10,6))
bars1=plt.bar(x-width/2,approved_votes,width,label="Approved")
bars2=plt.bar(x+width/2,rejected_votes,width,label="Rejected")
plt.title("KNN Loan Approval - Distance Measure Comparison")
plt.xlabel("Distance Measure")
plt.ylabel("Number of Votes")
plt.xticks(x,distance_measures)
plt.yticks([0,1,2,3])
plt.ylim(0,3.5)

for bar in bars1:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
for bar in bars2:
    height=bar.get_height()
    plt.text(bar.get_x()+bar.get_width()/2,height+0.05,str(int(height)),ha="center")
plt.legend()
plt.grid(axis="y",linestyle="--",alpha=0.4)
plt.tight_layout()
plt.show()
```



Question 10: Employee Attrition Prediction

A company wants to predict whether an employee is likely to Stay or Leave.

Employee	Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict	Class
E1	2	8	0	1	0	Stay
E2	3	7	0	1	0	Stay
E3	6	3	1	0	1	Leave
E4	7	2	1	0	1	Leave
E5	4	6	0	0	0	Stay
E6	8	2	1	0	1	Leave

A new employee has:

Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict
5	4	1	0	1

Using KNN with $K = 3$, classify the employee as Stay or Leave using all four distance measures.

a) Identify the numerical and binary features used for KNN classification.

Numerical Features

1. Experience in Years
2. Satisfaction Score

Numerical Features = Experience, Satisfaction Score

Binary Features

1. Overtime
2. Promotion Received
3. Workplace Conflict

Binary Features = Overtime, Promotion, Workplace Conflict

Target Variable

Class = Stay / Leave

b) Calculate the **Euclidean distance** between the new sample and all training samples

10)

(b) Euclidean distance

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

$$d = \sqrt{(5-2)^2 + (4-8)^2 + (1-0)^2 + (0-1)^2 + (1-0)^2}$$

$$= 5.292$$

$$d = \sqrt{(5-3)^2 + (4-7)^2 + (1-0)^2 + (0-1)^2 + (1-0)^2}$$

$$= 4.000$$

$$d = \sqrt{(5-6)^2 + (4-3)^2 + (1-1)^2 + (1-1)^2}$$

$$= 1.414$$

$$d = \sqrt{(5-7)^2 + (4-2)^2 + (1-1)^2 + (1-1)^2}$$

$$= 2.828$$

$$d = \sqrt{(5-4)^2 + (4-6)^2 + (1-0)^2 + (1-0)^2}$$

$$= 2.646$$

$$d = \sqrt{(5-8)^2 + (4-2)^2 + (1-1)^2 + (1-1)^2}$$

$$= 3.606$$

c) Calculate the **Manhattan distance** between the new sample and all training samples.

c) manhattan distance

$$d(x, y) = \sum |x_i - y_i|$$

$$3 + 4 + 1 + 1 + 1 = 10$$

$$2 + 3 + 1 + 1 + 1 = 8$$

$$1 + 1 = 2$$

$$2 + 2 = 4$$

$$1 + 2 + 1 + 1 = 5$$

$$3 + 2 + 0 = 5$$

d) Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.

(d) Minkowski Distance with $p=3$

$$d(x, y) = (\sum (x_i - y_i)^3)^{1/3}$$

$$E1: [(3)^3 + (4)^3 + (1)^3 + (1)^3 + (1)^3]^{1/3}$$
$$= 4.549$$

$$E2: [(2)^3 + (3)^3 + (1)^3 + (1)^3 + (1)^3]^{1/3}$$
$$= 3.362$$

$$E3: [(1)^3 + (1)^3]^{1/3}$$
$$= 1.260$$

$$E4: [(2)^3 + (2)^3]^{1/3}$$
$$= 2.520$$

$$E5: [(1)^3 + (2)^3 + (1)^3 + (1)^3]^{1/3}$$
$$= 2.224$$

$$E6: [(3)^3 + (2)^3 + 0]^{1/3}$$
$$= 3.271$$

e) Calculate the **Hamming distance** between the new sample and all training samples using binary features.

For Hamming distance, use only the binary features:

- Overtime
- Promotion Received

- Workplace Conflict

New employee:

$X=(1,0,1)$

Calculations

Employee	Binary Features	Hamming Distance	Class
E1	(0,1,0)	3	Stay
E2	(0,1,0)	3	Stay
E3	(1,0,1)	0	Leave
E4	(1,0,1)	0	Leave
E5	(0,0,0)	2	Stay
E6	(1,0,1)	0	Leave

f) Identify the **3 nearest neighbours** separately for each distance measure.

Distance Measure 1st Neighbour 2nd Neighbour 3rd Neighbour

Euclidean E3 (1.414) E5 (2.646) E4 (2.828)

Manhattan E3 (2) E4 (4) E5 (5)

Minkowski $p=3$ E3 (1.260) E5 (2.224) E4 (2.520)

Hamming E3 (0) E4 (0) E6 (0)

g) Predict the class label using majority voting for each distance measure.

Euclidean

Nearest neighbours:

- E3 → Leave
- E5 → Stay
- E4 → Leave

Leave=2, Stay=1 Euclidean = LEAVE

Manhattan

Nearest neighbours:

- E3 → Leave
- E4 → Leave
- E5 → Stay

Leave=2, Stay=1 Manhattan = LEAVE

Minkowski p=3

Nearest neighbours:

- E3 → Leave
- E5 → Stay
- E4 → Leave

Leave=2, Stay=1 Minkowski = LEAVE

Hamming

Nearest neighbours:

- E3 → Leave
- E4 → Leave
- E6 → Leave

Leave=3, Stay=0 Hamming = LEAVE

h) Compare the results and explain which distance measure is most suitable for the given scenario.

```
import matplotlib.pyplot as plt
```

```
distance_measures = ["Euclidean", "Manhattan", "Minkowski (p=3)", "Hamming"]
```

```
leave_votes = [2, 2, 2, 3]
```

```
stay_votes = [1, 1, 1, 0]
```

```
plt.figure(figsize=(10, 6))
```

```
plt.plot(distance_measures, leave_votes, marker="o", linewidth=2, label="Leave")
```

```
plt.plot(distance_measures, stay_votes, marker="o", linewidth=2, label="Stay")
```

```
plt.title("Employee Attrition KNN - Distance Measure Comparison")
```

```
plt.xlabel("Distance Measure")
```

```
plt.ylabel("Majority Votes")
```

```
plt.ylim(0, 3.5)
```

```
plt.yticks([0, 1, 2, 3])
```

```
for i, value in enumerate(leave_votes):  
    plt.text(i, value + 0.08, str(value), ha="center")
```

```
for i, value in enumerate(stay_votes):  
    plt.text(i, value + 0.08, str(value), ha="center")  
plt.legend()  
plt.grid(True, linestyle="--", alpha=0.4)  
plt.tight_layout()  
plt.show()
```

